Quantitative Methods in Finance

Final Exam — Take-Home

Counterfactual Inflation Analysis:

What If Ukraine Had Been Part of the Euro Area?

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Context and Motivation

On 23 June 2022, the European Council granted Ukraine official candidate status for accession to the European Union. While EU membership and Euro Area membership are distinct steps—separated by years of institutional convergence and the fulfilment of the Maastricht criteria—the question of whether Ukraine could eventually adopt the euro is no longer purely hypothetical. It is a question that central banks, including the ECB and the national central banks of the Eurosystem, will increasingly need to assess.

This exam asks you to engage with a foundational question at the intersection of monetary economics and applied econometrics:

What would Ukraine's inflation trajectory have looked like had it been a member of the Euro Area?

This is a **counterfactual** question. Ukraine was never in the Euro Area; it maintained its own currency (the hryvnia), its own central bank (the National Bank of Ukraine, NBU), and its own monetary policy throughout the sample period. Your task is to construct a plausible alternative inflation path for Ukraine under the hypothetical scenario of Euro Area membership, using the econometric methods studied in this course.

Why This Question Matters

The academic literature on **Optimal Currency Areas** (OCA), initiated by Mundell (1961) and extended by McKinnon (1963) and Kenen (1969), identifies the fundamental trade-off faced by a country considering membership in a monetary union: the *loss of independent monetary and exchange-rate policy* versus the *gains from reduced transaction costs, price transparency, and imported monetary credibility.*

For Ukraine, this trade-off is particularly stark. Consider the following channels through which Euro Area membership would have reshaped the inflation process:

1. **Loss of the exchange-rate buffer.** Ukraine experienced three major hryvnia devaluations: approximately −50% against the US dollar in 2008–2009 (global financial crisis), over −200% in 2014–2015 (Crimea annexation and Donbas conflict), and approximately −25% in 2022 (full-scale invasion). These devaluations were a critical adjustment mechanism. Under Euro Area membership, this channel would have been *unavailable*. The real adjustment would have had to occur through internal devaluation—wage and price deflation—as experienced by Greece, Ireland, and the Baltic states during the 2010–2012 sovereign debt crisis.

2. **Imported monetary credibility.** The ECB's inflation-targeting framework and institutional independence provide a nominal anchor that the NBU has historically lacked. Before the adoption of inflation targeting in 2016, Ukraine operated under various exchange-rate and monetary-aggregate regimes with limited credibility, leading to persistent inflation expectations well above Euro Area levels. Euro Area membership would have *imported* the ECB's credibility, likely anchoring inflation expectations at lower levels—the so-called Frankel and Rose (1998) endogeneity hypothesis, whereby joining a currency union makes a country *more* suitable for it ex post.

3. **One monetary policy for heterogeneous shocks.** The ECB sets a single policy rate for the entire Euro Area. Ukraine's economy is structurally different from the Euro Area core: it is more exposed to energy-price shocks (historically dependent on Russian natural gas), more reliant on agricultural commodity exports, and subject to geopolitical shocks that are largely asymmetric—i.e., they do not affect Euro Area members similarly. A single monetary policy calibrated for the Euro Area aggregate might not have been *optimal* for Ukraine's specific conditions.

4. **Fiscal discipline and the Stability and Growth Pact.** Euro Area membership comes with fiscal constraints (the 3% deficit and 60% debt-to-GDP rules, however imperfectly enforced). These constraints would have limited Ukraine's ability to use fiscal policy as a substitute for the lost monetary-policy and exchange-rate instruments—amplifying the cost of asymmetric shocks but potentially reducing the inflation bias associated with deficit monetisation.

5. **Financial integration and capital flows.** Euro Area membership would have eliminated currency risk for foreign investors, likely resulting in lower sovereign borrowing costs and larger capital inflows *in normal times*—but also removing the ability to impose capital controls or let the exchange rate depreciate during sudden stops, as Ukraine did in 2008–2009, 2014–2015, and 2022.

The counterfactual you construct is therefore not merely a statistical exercise. It embodies a set of **structural hypotheses** about how the inflation process would have differed under an alternative monetary regime. The quality of your answer depends on how explicitly and coherently you operationalise these hypotheses in your econometric framework.

Part A — Ukraine's Monetary Regime: A Preliminary Analysis (30%)

Before constructing a counterfactual, you must understand *what Ukraine actually did* with its monetary sovereignty. A naïve framing assumes Ukraine operated under a freely floating exchange rate with full monetary policy independence throughout the sample period. This is incorrect.

Task

Construct a **documented chronology** of the National Bank of Ukraine's exchange rate regime from 2000 to 2025. Your chronology must identify:

(a) The periods during which the NBU maintained a **de facto peg** to the US dollar (and the approximate level of the peg);

(b) The **major devaluation episodes**, with approximate dates, magnitudes, and triggers;

(c) The periods during which significant **capital controls** were in place;

(d) The transition to **inflation targeting** (official date, target path, and whether the NBU's behaviour actually changed in practice);

(e) For each period, your assessment of the **IMF's de facto classification** of Ukraine's exchange rate arrangement (stabilised arrangement, managed float, conventional peg, etc.).

Deliverables for Part A

1. A **summary table** with columns: period, approximate UAH/USD rate, de facto regime classification, and key events.
2. A **brief argument** (one paragraph, in your code or a separate text file) answering: *During which periods did Ukraine have genuine monetary sovereignty? During which periods was its monetary policy de facto constrained by the exchange-rate peg? What does this imply for the "treatment" that Euro Area membership would represent?*

Why this matters

This preliminary analysis is not optional background—it is central to your identification strategy in Part B. If Ukraine was already operating a de facto dollar peg for most of the sample, then Euro Area membership would not have represented a large change in monetary regime during the peg periods. The "treatment" of joining the Euro Area is concentrated in the episodes where Ukraine *deviated* from its peg—the devaluations and the post-2016 inflation-targeting period. Your counterfactual must reflect this asymmetry.

Sources. You should consult IMF as well as NBU publications and academic sources. Cite your sources.

Data

Repository data

The following datasets are available in the GitHub repository:

https://github.com/skimeur/pioneer-detection-method

- `data_ecb_hicp_panel.csv` — Monthly Harmonised Index of Consumer Prices (HICP), year-on-year percentage change, for 11 Euro Area countries (Austria, Belgium, Germany, Spain, Finland, France, Greece, Ireland, Italy, Netherlands, Portugal), from January 2000 to December 2025. Source: ECB Data Portal.
  - Nota bene: you might need to augment this list of countries if you do a synthetic control.
- `data_ukraine_cpi_raw.csv` — Monthly Consumer Price Index for Ukraine, expressed as an **index with the previous month as base** (i.e., a value of 101.5 means prices rose 1.5% month-on-month), from February 2005 to December 2025. Source: State Statistics Service of Ukraine (SSSU) via SDMX.

Important note on data harmonisation. The two datasets are *not* in the same format. The Euro Area data are year-on-year inflation rates (%), while the Ukrainian data are month-on-month price indices. You will need to transform the Ukrainian series into year-on-year inflation rates to make it comparable with the Euro Area panel.

External data: a mandatory requirement

You are required to augment the repository data with additional macroeconomic time series from external sources. The methods studied in this course—structural VARs,

local projections—require variables beyond inflation alone. At a minimum, you should consider sourcing:

- **Real GDP or industrial production** for Ukraine and Euro Area countries (quarterly or monthly), from Eurostat, the IMF's International Financial Statistics (IFS), or the World Bank;
- **Exchange rate series** (UAH/USD, UAH/EUR) from the NBU or IMF;
- Any other macroeconomic variables your identification strategy requires (policy rates, commodity prices, fiscal balances, etc.).

The quality, relevance, and documentation of your external data choices are **part of the grading**. You must cite your data sources and explain why each variable is needed for your econometric framework.

Key Literature for the Counterfactual

The following papers are selected not as a general reading list, but as **direct methodological inputs** to the construction of your counterfactual. Each one offers a specific tool, identification strategy, or conceptual framework that you can adapt. You are not required to read all of them—choose those relevant to your chosen approach.

Structural shock decomposition: Bayoumi and Eichengreen (1993)

Bayoumi and Eichengreen (1993) estimate bivariate structural VARs (output growth, inflation) for pairs of countries, using the Blanchard and Quah (1989) long-run identification: supply shocks have permanent effects on output, demand shocks do not. They then compare the *correlation of structural shocks* across countries to assess OCA suitability, distinguishing a European "core" (whose shocks are symmetric) from a "periphery" (whose shocks are idiosyncratic).

How to use it for the counterfactual. You can replicate the Bayoumi–Eichengreen decomposition for Ukraine and each Euro Area country, then construct the counterfactual via *shock replacement*: if Ukraine had been in the Euro Area, its **demand shocks** would have been determined by ECB monetary policy (i.e., the Euro Area aggregate demand shocks), while its **supply shocks**—driven by real factors (energy prices, agriculture, geopolitics)—would have remained its own. The counterfactual inflation path is obtained by feeding Ukraine's supply shocks and the Euro Area's demand shocks through Ukraine's (or the Euro Area's) estimated impulse response functions. This requires both output and inflation data, which is why external data sourcing is mandatory.

Common inflation factors: Ciccarelli and Mojon (2010)

Ciccarelli and Mojon (2010) show that a large fraction of inflation variation across countries can be explained by a single common factor, which they interpret as "global inflation." Their approach extracts a common component from a panel of inflation series using dynamic factor model techniques (principal components or state-space estimation).

How to use it for the counterfactual. Estimate the Euro Area common inflation factor from the panel of 11 countries. Then model Ukraine's inflation as:

$$\pi_{UKR,t} \;=\; \lambda_{UKR}\, F_t^{EA} \;+\; \varepsilon_{UKR,t}$$

where F_t^{EA} is the Euro Area common factor, λ_{UKR} is Ukraine's loading on that factor (estimated from "quiet" periods when Ukraine's inflation co-moved with Europe), and $\varepsilon_{UKR,t}$ is the

Ukraine-specific residual. The counterfactual "Ukraine-in-the-Euro-Area" inflation is $\hat{\pi}^{CF}_{UKR,t} = \hat{\lambda}_{UKR}\,\hat{F}^{EA}_t$: only the common European component, without the idiosyncratic shocks that Ukraine's own monetary policy and exchange rate produced. This is a panel method that can be implemented with the inflation data alone; adding macroeconomic covariates strengthens identification.

Synthetic control: Abadie, Diamond, and Hainmueller (2010)

Abadie et al. (2010) propose constructing a "synthetic" version of a treated unit as a convex combination of untreated donor units, with weights chosen to match the treated unit's pre-treatment trajectory.

A difficulty for this exercise. In the classic synthetic control setup, there is a clean treatment date (before which the treated unit and its synthetic counterpart should track each other, and after which they diverge). For Ukraine, **there is no clean treatment date**: Ukraine never joined the Euro Area, so there is no before/after. You may attempt to invert the logic—treating Ukraine's *absence* from the Euro Area as the treatment—but this requires careful conceptual justification.

A note on recent Euro Area accessions. Since 1999, ten countries have adopted the euro. Several of them—particularly among the post-Soviet Baltic states—share structural features with Ukraine (small, open, post-Soviet economies with a history of exchange-rate pegs). Their accession dates and pre/post inflation paths may contain useful information for your analysis:

Country	EU accession	Euro adoption	Notes
Greece	1981	1 Jan 2001	Debt crisis 2010–2015
Slovenia	2004	1 Jan 2007	First 2004-enlargement country
Cyprus	2004	1 Jan 2008	Banking crisis 2012–2013
Malta	2004	1 Jan 2008	Small island economy
Slovakia	2004	1 Jan 2009	Adopted euro during GFC
Estonia	2004	1 Jan 2011	Post-Soviet; currency board pre-euro
Latvia	2004	1 Jan 2014	Post-Soviet; severe 2008–09 recession
Lithuania	2004	1 Jan 2015	Post-Soviet; last Baltic to join
Croatia	2013	1 Jan 2023	De facto euro-linked kuna pre-adoption
Bulgaria	2007	1 Jan 2026	Currency board since 1997

Constrained sovereignty: Calvo and Reinhart (2002)

Calvo and Reinhart (2002) document that many countries that officially declare a floating exchange rate in practice intervene heavily to limit exchange-rate movements—a pattern they call "fear of floating." The implication is that *de jure* monetary sovereignty can be very different from *de facto* monetary sovereignty.

Why this is critical for Ukraine. As your Part A analysis should reveal, the NBU maintained a de facto peg to the US dollar for most of the period 2000–2014, and returned to a fixed rate under wartime conditions in 2022–2023. During peg periods, Ukraine had *no independent monetary policy* in practice—the "impossible trinity" was binding. This means that the counterfactual "treatment" of Euro Area membership is **not constant over time**: it is large during episodes when Ukraine exercised genuine monetary autonomy (the devaluations, the post-2016 inflation-targeting period) and small during peg periods (when monetary policy was already subordinated to the exchange-rate anchor). Your identification strategy should account for this time-varying treatment intensity.

Credibility import: Giavazzi and Pagano (1988); Barro and Gordon (1983)

Giavazzi and Pagano (1988) argue that countries with weak monetary credibility can "tie their hands" by joining a fixed exchange-rate arrangement (or a monetary union), importing the credibility of the anchor country's central bank. Barro and Gordon (1983) provide the theoretical foundation: under discretion, a central bank faces a time-consistency problem that generates an inflation bias; an external commitment device eliminates this bias.

Relevance for Ukraine. The NBU's credibility was low before 2016 (no formal inflation target, repeated peg collapses). After adopting inflation targeting in 2016, credibility improved but remained fragile. A **sanity check** for your counterfactual: the gap between actual Ukrainian inflation and your counterfactual Euro Area path should be *larger* in the pre-2016 period (when credibility gains from Euro Area membership would have been substantial) and potentially *smaller* after 2016 (when the NBU had partially converged toward a modern inflation-targeting framework). If your counterfactual does not exhibit this pattern, reconsider your assumptions.

Sovereign fragility in monetary unions: De Grauwe (2012)

De Grauwe (2012) argues that members of a monetary union are uniquely vulnerable to self-fulfilling sovereign debt crises, because they cannot rely on their own central bank as a lender of last resort in the currency of denomination. This fragility was illustrated by the sovereign spreads divergence in 2010–2012 and the eventual need for the ECB's Outright Monetary Transactions (OMT) programme.

Relevance for Ukraine. Had Ukraine been in the Euro Area during the 2014 or 2022 crises, it would have avoided a *currency crisis* but might have faced a *sovereign debt crisis* instead—with no national central bank to provide emergency liquidity. The counterfactual is therefore not simply "lower inflation" but potentially "a different type of macroeconomic crisis." This is not a required modelling element, but acknowledging this channel in your interpretation will demonstrate deeper understanding of the trade-offs.

Part B — The Counterfactual (50%)

Using the methods studied in this course—and any additional econometric tools you deem appropriate—construct a counterfactual inflation time series for Ukraine under the hypothesis that it had been a member of the Euro Area.

Deliverables for Part B

1. **A single figure** showing two time series on the same axes:
   - Ukraine's *actual* year-on-year inflation (computed from the raw data);
   - Your *counterfactual* "Ukraine-in-the-Euro-Area" inflation.

   The figure should be clearly labelled, with a legend, axis titles, and appropriate date formatting.
2. **A brief interpretation** (one paragraph) answering: *What does your counterfactual imply about the cost or benefit of monetary sovereignty for Ukraine, particularly during the 2008–2009, 2014–2015, and 2022 crises?*
3. **Your code**, which must be fully reproducible. All external data must either be downloaded programmatically within your code or provided alongside it with clear documentation.

Methodological requirements

- Your method must reflect an **explicit identification strategy**: you must articulate (through your code structure and comments) what "being in the Euro Area" means in terms of your model's parameters, shocks, or constraints.
- **Your counterfactual must differ from the simple cross-sectional mean of Euro Area countries' inflation.** This constraint ensures that your approach involves genuine econometric reasoning rather than a mechanical average.
- Your identification strategy must be **consistent with your Part A analysis**. If you found that Ukraine had no genuine monetary sovereignty during peg periods, your counterfactual should reflect a smaller "treatment effect" during those periods.
- You are expected to use **structural VARs, local projections, or comparable methods** studied in this course. You may complement them with panel methods (factor models, synthetic control), but the core of your analysis should demonstrate mastery of the SVAR/LP toolkit.

Grading Criteria

Component	Weight	Criteria
Part A: Monetary regime	30%	Accuracy of the chronology; quality of sources; coherence of the argument about which periods represent genuine monetary sovereignty; correct identification of de facto vs. de jure regimes.
Part B: Identification	30%	Is the counterfactual grounded in a coherent economic and econometric framework? Are the assumptions explicit? Does the method go beyond mechanical projection? Is it consistent with Part A?
Part B: Technical execution	20%	Is the code correct, reproducible, and well-structured? Is the data transformation handled properly? Are estimation choices (lag selection, stationarity testing, external data sourcing) justified?
Part B: Output and interpretation	20%	Is the figure informative and professionally presented? Does the counterfactual exhibit plausible dynamics? Does the interpretation demonstrate economic understanding of the OCA trade-offs?

Practical Information

- **Format:** take-home exam, individual work.
- **Duration:** you have until **1st of May 2026** to submit.
- **Submission:** upload your code, output figure(s), and Part A deliverables to **GitHub**.
- **Language:** any programming language (Python, R, Julia, MATLAB, ...). The code must run from the repository data and your documented external data without manual intervention.
- **External data:** all external datasets must be either (a) downloaded programmatically within your code, or (b) provided as files alongside your submission with full source documentation.
- **Collaboration policy:** this is individual work. You may consult any published reference, documentation, or AI tool, but the econometric reasoning and code must be your own.

Identical or near-identical submissions will be treated as plagiarism.

Suggested References

The following references supplement those discussed in the literature section above. Choose those relevant to your chosen method.

Optimal Currency Areas
Mundell (1961); McKinnon (1963); Kenen (1969); Frankel and Rose (1998); Alesina and Barro (2002).

Structural VARs and local projections
Sims (1980); Blanchard and Quah (1989); Stock and Watson (2001); Jordà (2005); Kilian and Lütkepohl (2017).

Counterfactual methods
Abadie et al. (2010) (synthetic control); Angrist and Pischke (2009) (DiD and potential outcomes).

Euro Area inflation dynamics
Angeloni and Ehrmann (2007); Ciccarelli and Mojon (2010).

Monetary credibility and exchange-rate regimes
Barro and Gordon (1983); Giavazzi and Pagano (1988); Calvo and Reinhart (2002); De Grauwe (2012).

Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.

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Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

Barro, R. J. and Gordon, D. B. (1983). Rules, discretion and reputation in a model of monetary policy. *Journal of Monetary Economics*, 12(1):101–121.

Bayoumi, T. and Eichengreen, B. (1993). Shocking aspects of European monetary integration. In Torres, F. and Giavazzi, F., editors, *Adjustment and Growth in the European Monetary Union*, pages 193–229. Cambridge University Press.

Blanchard, O. J. and Quah, D. (1989). The dynamic effects of aggregate demand and supply disturbances. *American Economic Review*, 79(4):655–673.

Calvo, G. A. and Reinhart, C. M. (2002). Fear of floating. *The Quarterly Journal of Economics*, 117(2):379–408.

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Frankel, J. A. and Rose, A. K. (1998). The endogeneity of the optimum currency area criteria. *The Economic Journal*, 108(449):1009–1025.

Giavazzi, F. and Pagano, M. (1988). The advantage of tying one's hands: EMS discipline and central bank credibility. *European Economic Review*, 32(5):1055–1075.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

Kenen, P. B. (1969). The theory of optimum currency areas: An eclectic view. In Mundell, R. A. and Swoboda, A. K., editors, *Monetary Problems of the International Economy*, pages 41–60. University of Chicago Press.

Kilian, L. and Lütkepohl, H. (2017). *Structural Vector Autoregressive Analysis*. Cambridge University Press.

McKinnon, R. I. (1963). Optimum currency areas. *American Economic Review*, 53(4):717–725.

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Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1):1–48.

Stock, J. H. and Watson, M. W. (2001). Vector autoregressions. *Journal of Economic Perspectives*, 15(4):101–115.